From the above equations,

5.

Particle speeds on entering the magnetic field

At open . Since , 's voltage is , so all particles pass . After , if they reach at or integer multiples of , 's voltage is zero, so they pass . To pass then ,

their speeds must satisfy

For particles passing at then , speeds are

Time from to at speed :

Time from to :

Thus arrival time at is

Since shuts at , only if

they pass into the field. Using , is an integer, so the largest is

Their energies on entering the magnetic field are

Their energies are, respectively,